National Aquatic Connectivity Collaborative

AQUATIC CONNECTIVITY

Stream Crossing Survey

DATA FORM

Database entry by _____ Entry date _____

Data entry reviewed by _____ Review date _____

CROSSING DATA

Crossing Code _____ **Local ID** (Optional) _____

Lead Observer _____ **Date Observed** (MM/DD/YYYY) _____

Stream _____ **Road** _____

Tidal Site ☐ YES ☐ NO ☐ UNKNOWN **Type** ☐ MULTILANE ☐ PAVED ☐ UNPAVED ☐ DRIVEWAY ☐ TRAIL ☐ RAILROAD

Repeat Survey ☐ YES ☐ NO ☐ UNKNOWN **Past Project** (for aquatic connectivity) ☐ COMPLETED ☐ PROPOSED ☐ UNKNOWN **County** (or Town, optional) _____

GPS Coordinates (Decimal degrees) ___ . ___ °N Latitude — ___ . ___ °W Longitude **Location Description** _____

Crossing Type ☐ BRIDGE ☐ CULVERT ☐ MULTIPLE CULVERT ☐ FORD ☐ VENTED FORD ☐ NO CROSSING ☐ BRIDGE ADEQUATE ☐ REMOVED CROSSING ☐ BURIED STREAM ☐ INACCESSIBLE ☐ PARTIALLY INACCESSIBLE ☐ NO UPSTREAM CHANNEL **Number of Culverts / Bridge Cells** _____

Structure Owner Type ☐ FEDERAL ☐ STATE ☐ LOCAL ☐ PRIVATE ☐ PUBLIC UTILITY ☐ OTHER ☐ UNKNOWN **Road Fill Height** (Top of culvert to road surface; bridge=0) _____

Flow Condition ☐ NO FLOW ☐ TYPICAL-LOW ☐ MODERATE ☐ HIGH **Crossing Condition** ☐ NEW ☐ OK ☐ POOR ☐ FAILING ☐ UNKNOWN

Stream Width _____ **Width based on** ☐ ACTIVE CHANNEL ☐ WETTED CHANNEL ☐ BANKFULL WIDTH **Confidence** ☐ HIGH ☐ LOW / ESTIMATED

Constriction ☐ SEVERE ☐ MODERATE ☐ SPANS ONLY BANKFULL/ ACTIVE CHANNEL ☐ SPANS FULL CHANNEL & BANKS **Alignment** ☐ FLOW-ALIGNED ☐ SKEWED (>45°)

Tailwater Scour Pool ☐ NONE ☐ SMALL ☐ LARGE **Tailwater Riparian Vegetation** (optional) **Ground level** ☐ HIGH (>50%) ☐ LOW **Understory** ☐ HIGH ☐ LOW **Overstory** ☐ HIGH ☐ LOW

Inlet Scour Pool ☐ NONE ☐ SMALL ☐ LARGE **Inlet Riparian Vegetation** (optional) **Ground level** ☐ HIGH (>50%) ☐ LOW **Understory** ☐ HIGH ☐ LOW **Overstory** ☐ HIGH ☐ LOW

Photo IDs Inlet _____ Outlet _____ Upstream _____ Downstream _____ Other _____

Crossing Comments _____ **BATS PRESENT?** ☐ YES (number unknown) ☐ FEW ☐ MANY (≥ 10)

STRUCTURE 1

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** ___ . ___ **Bottom of structure** ___ . ___

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN **Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN **Undermining** ☐ YES ☐ NO

Dimensions A. Width ___ . ___ B. Height ___ . ___ C. Substrate/Water Width ___ . ___ D. Water Depth ___ . ___ **Velocity** (Optional) ___ . ___ m/s

E. Abutment Height (Type 7 bridges only) ___ . ___ **Drop To Water Surface** ___ . ___ **Drop To Stream Bottom** ___ . ___ **Temperature** (Optional) ___ . ___ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN **Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN **Undermining** ☐ YES ☐ NO

Dimensions A. Width ___ . ___ B. Height ___ . ___ C. Substrate/Water Width ___ . ___ D. Water Depth ___ . ___ **Velocity** (Optional) ___ . ___ m/s

Drop To Water Surface ___ . ___ **Drop To Stream Bottom** ___ . ___ **Temperature** (Optional) ___ . ___ °F

ADDITIONAL CONDITIONS

Slope (Optional) ___ % **Slope Confidence** ☐ HIGH ☐ LOW **Internal Structures** ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments _____

STRUCTURE 2

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments

STRUCTURE 3

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments

STRUCTURE 4

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments

STRUCTURE 5

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments

STRUCTURE 6

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

Comments

STRUCTURE 7

Material ☐ METAL ☐ CONCRETE ☐ PLASTIC ☐ WOOD ☐ ROCK/STONE ☐ FIBERGLASS ☐ COMBINATION

L. Structure Length **Top of structure** _____.__ **Bottom of structure** _____.__

OUTLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Grade ☐ AT STREAM GRADE ☐ FREE FALL ☐ CASCADE ☐ FREE FALL ONTO CASCADE ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

E. Abutment Height (Type 7 bridges only) _____.__

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

INLET

Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN

Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Type ☐ PROJECTING ☐ HEADWALL ☐ WINGWALLS ☐ HEADWALL & WINGWALLS ☐ MITERED TO SLOPE ☐ OTHER ☐ NONE

Grade ☐ AT STREAM GRADE ☐ INLET DROP ☐ PERCHED ☐ CLOGGED/COLLAPSED/SUBMERGED ☐ UNKNOWN

Undermining ☐ YES ☐ NO

Dimensions **A. Width** _____.__ **B. Height** _____.__ **C. Substrate/Water Width** _____.__ **D. Water Depth** _____.__

Velocity (Optional) _____.__ m/s

Drop To Water Surface _____.__ **Drop To Stream Bottom** _____.__

Temperature (Optional) _____.__ °F

ADDITIONAL CONDITIONS

Slope (Optional) _____ % **Slope Confidence** ☐ HIGH ☐ LOW

Internal Structures ☐ NONE ☐ BAFFLES/WEIRS ☐ SUPPORTS ☐ OTHER _____

Structure Substrate **Matches Stream** ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE ☐ UNKNOWN

Type (Pick one) ☐ NONE ☐ SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ ORGANIC MATERIAL ☐ UNKNOWN

Coverage ☐ NONE ☐ 25% ☐ 50% ☐ 75% ☐ 100% ☐ UNKNOWN

Physical Barriers (Pick all that apply) ☐ NONE ☐ DEBRIS/SEDIMENT/ROCK ☐ DEFORMATION ☐ FREE FALL ☐ FENCING ☐ DRY ☐ OTHER

Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE

Depth Matches Stream ☐ YES ☐ SHALLOWER ☐ DEEPER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Velocity Matches Stream ☐ YES ☐ FASTER ☐ SLOWER ☐ DRY ☐ DRY-STREAM ALSO DRY ☐ UNKNOWN

Dry Passage through Structure? ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage** _____

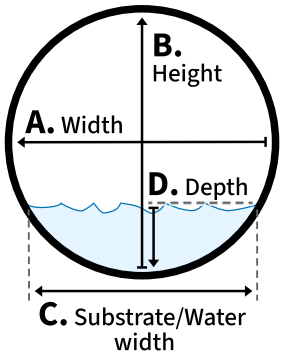
Comments

Structure Shape & Dimensions

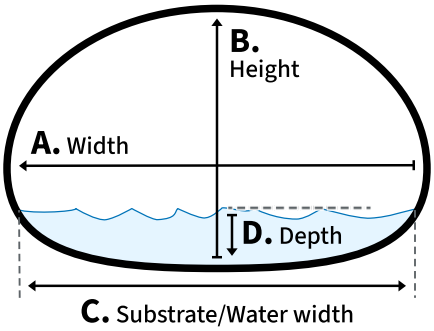
1) Select the Structure Shape number from the following diagrams and record it on the form for Inlet and Outlet Shape.
2) Record on the form in the approriate blanks dimensions **A**, **B**, **C**, and **D** as shown in the diagrams; **C** captures the width of water or substrate, whichever is wider; for dry culverts without substrate, C = 0.
   D is the depth of water -- be sure to measure inside the structure; for dry culverts, D = 0.
3) Record Structure Length (**L**). (Record abutment height (**E**) only for Type 7 Structures.)
4) For multiple culverts, also record the Inlet and Outlet shape and dimensions for each additional culvert.

NOTE: Culverts 1, 2 & 4 may or may not have substrate in them, so height measurements (**B**) are taken from the level of the "stream bed", whether that bed is composed of substrate or just the inside bottom surface of a culvert.

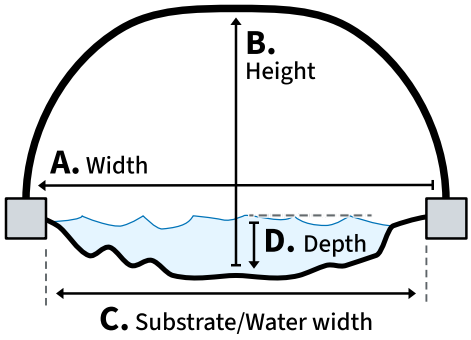
1. Round Culvert



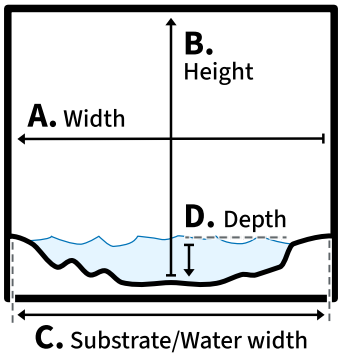
2. Pipe Arch / Elliptical Culvert



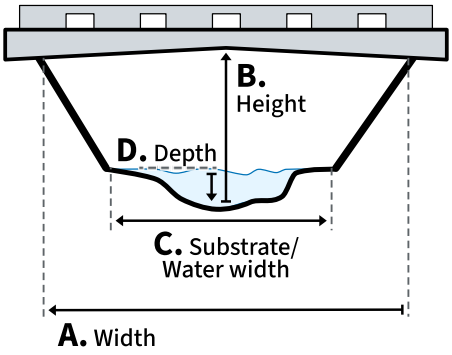
3. Open Bottom Arch Bridge / Culvert



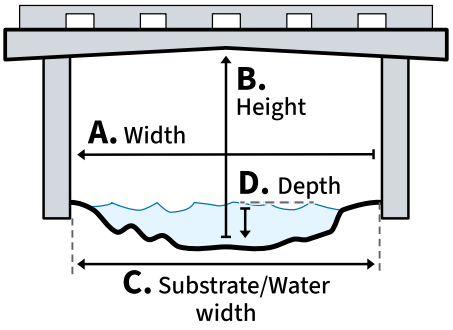
4. Box Culvert



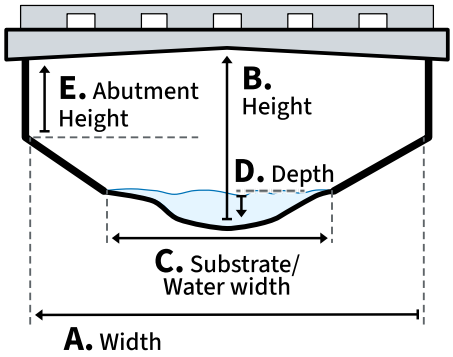
5. Bridge with Side Slopes



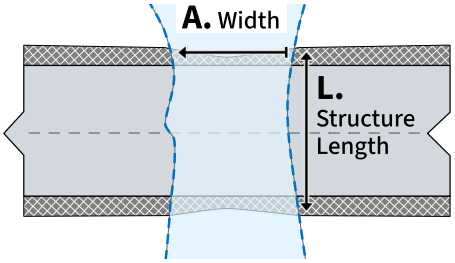
6. Box / Bridge with Abutments

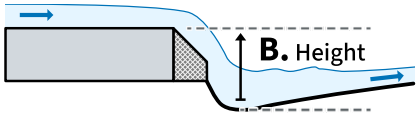


7. Bridge with Side Slopes & Abutments

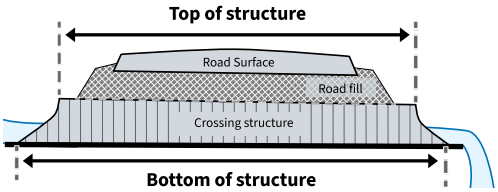


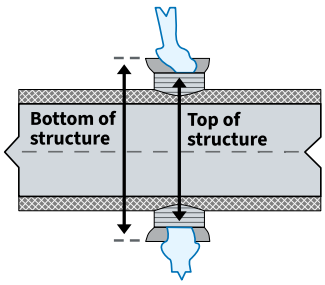
Ford





L. Structure Length

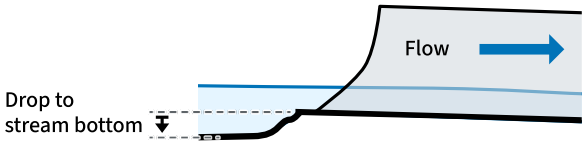




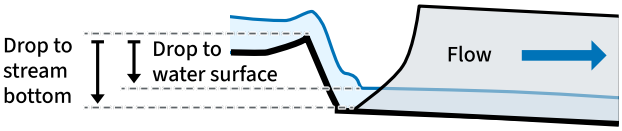
Inlet Drop Measurements

Record the Inlet Drop to Water Surface and Inlet Drop to Stream Bottom as indicated in the following diagrams for the Inlet Grade.

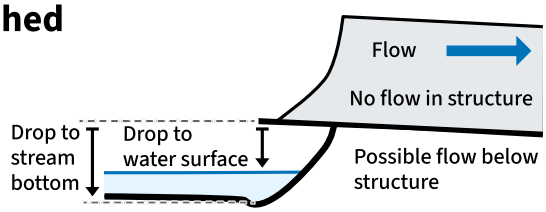
At Stream Grade



Inlet drop



Perched

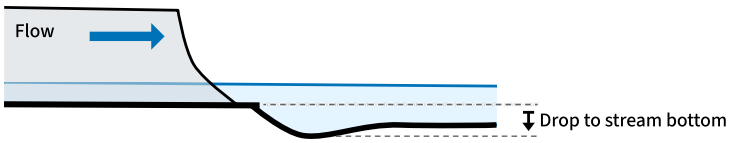


Outlet Drop Measurements

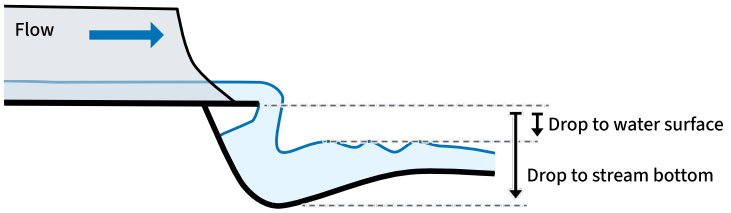
Record the Outlet Drop to Water Surface and Outlet Drop to Stream Bottom as indicated in the following diagrams for the Outlet Grade.

NOTE: For Cascade and Free Fall Onto Cascade, measure Outlet Drop to Stream Bottom at the end of the cascade.

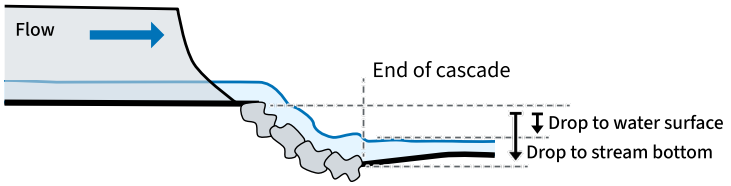
At Stream Grade



Free Fall



Cascade



Free Fall Onto Cascade

